

Exam policy:

- *The exam is 50 minutes (9:00–9:50 AM) and is closed book, closed notes.*
- *You are allowed one hand-written cheat sheet of at most one side of one page. Please hand in your cheat sheet with your solutions.*
- *Justify all your answers. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.*
- *No calculators.*

Each of the three questions is worth an equal number of points.

The exam questions are on the other side of this page.

Please do not turn over this page until you are told to start the exam.

Good luck!

Q. 1 (Questions on the exam). The exam has three questions. They are in increasing order of difficulty: you get the first question right with probability $\frac{3}{4}$; the second question right with probability $\frac{1}{2}$; and the third question right with probability $\frac{1}{4}$. Your success in answering each question is independent.

- a. What is the expected number of questions you get right?
- b. What is the variance of the number of questions you get right?
- c. Given that you got two out of three questions right, what is the probability that you answered question number three correctly?

Q. 2 (Dice). You repeatedly and independently throw a fair die. What is:

- a. the probability that you do not throw a 6 in the first k throws?
- b. the expected number of throws until you throw a 6 for the first time?
- c. the probability that you do not throw any 1 or 2 before the first time you throw a 6?

Hint: condition on the event that you throw a 6 for the first time in the k th throw.

Q. 3 (Ice cream). The craving for ice cream can come at any time. We model the first time (starting now, in hours) at which you crave ice cream as an exponentially distributed random variable with a rate parameter of 1 (craving per hour).

- a. What is the probability that the craving has not yet come in the first hour?
- b. Suppose the craving has not yet come in the first hour. Given this information, what is the probability that the craving comes during the second hour?
- c. The ice cream store designates the first 6 minutes of every hour “happy minutes” during which the ice cream goes on sale. What is the probability that the craving comes on exactly during one of these (infinitely many) “happy minutes” periods?